

SPOTIFY TRACKS ANALYSIS DATA-DRIVEN INSIGHTS ON LISTENS, ENGAGEMENT & AUDIO FEATURES

Analyzed using SQL the Spotify dataset and deliver actionable insights on tracks, artists and albums.



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PROBLEM STATEMENT

Spotify's content and marketing team aims to analyze:

- Which tracks and artists drive the highest engagement
- Differences in listener behavior across platforms (Spotify vs YouTube)
- High-potential tracks for promotion based on streams, views, and energy
- Album-level performance and consistency in audio characteristics
- Opportunities to improve recommendations and playlist strategies

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About our dataset

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[Find dataset here.](#)

Tables used:

- The dataset contains attributes such as:
- Artist: The performer of the track.
- Track: The name of the song.
- Album: The album to which the track belongs.
- Album_type: The type of album (e.g., single or album).
- Various metrics such as danceability, energy, loudness, tempo, and more.



Dataset



0:00

3:55



Yawar Ali

Leader

FOLLOW

SHUFFLE PLAY

Technical Skills Demonstrated

- SELECT Statements
- GROUP BY
- ORDER BY
- Aggregate Functions
- CASE Statements
- Joins
- Subqueries
- CTEs
- Window Functions
- Ranking Functions

SEE MORE

1

Streaming Performance Analytics



```
-- 1. Retrieve the names of all tracks that have more than 1 billion streams
Select * from spotify
where Stream > 1000000000
```

```
-- Answer: 385 songs
```

```
-- 2. List all albums along with their respective artists.
Select DISTINCT Album, Artist
from spotify
order by 1
```

```
-- Answer: 14,177 Distinct Albums with their artists
```

```
-- 3. Get the total number of comments for tracks where licensed = TRUE.
SELECT sum(Comments) AS CommentCount
FROM spotify
WHERE Licensed = 1
```

```
-- Answer: 497,023,475 Comments
```

```
-- 4. Find all tracks that belong to the album type single.
Select count(Album) as Singles
from spotify
where Album_type = 'single'
```

```
-- Answer: 4,973 total songs
```

```
-- 5. Count the total number of tracks by each artist.
Select Artist, count(*) as total_songs
from spotify
group by Artist
```

```
-- Answer: 2,074 total singers
```



2

Engagement & Trend Analytics



```
-- 1. Calculate the average danceability of tracks in each album.
SELECT Album,
       AVG(Danceability) AS avg_danceability
FROM spotify
GROUP BY Album
order by 2 desc
```

```
-- Answer: 11,799 albums calculated with top avg_danceability
-- FOR CERTAIN (Deluxe) 0.975000023841858
```

```
--5. Retrieve the track names that have been streamed on Spotify more than YouTube.
SELECT Track,
       SUM(CASE WHEN most_playedon = 'Youtube' THEN Stream ELSE 0 END) AS streamed_on_youtube,
       SUM(CASE WHEN most_playedon = 'Spotify' THEN Stream ELSE 0 END) AS streamed_on_spotify
FROM spotify
GROUP BY Track

SELECT *
FROM (
    SELECT Track,
           COALESCE(SUM(CASE WHEN most_playedon = 'Youtube' THEN Stream ELSE 0 END), 0) AS streamed_on_youtube,
           COALESCE(SUM(CASE WHEN most_playedon = 'Spotify' THEN Stream ELSE 0 END), 0) AS streamed_on_spotify
    FROM spotify
    GROUP BY Track
) AS t1
WHERE streamed_on_spotify > streamed_on_youtube
AND streamed_on_youtube <> 0

-- Answer: 185 songs performing well on Spotify than Youtube
```

```
-- 2. Find the top 5 tracks with the highest energy values.
SELECT TOP 5
       Track,
       MAX(EnergyLiveness) AS Max_EnergyLiveness
FROM spotify
GROUP BY Track
ORDER BY Max_EnergyLiveness DESC
```

```
-- Answer: Take It 59.1139259338379
-- Verano Azul 58
-- Salvavidas 57.6551742553711
-- Ants Marching 54.5222930908203
-- Eres Mi Sueño - Versión Radio Edit 51.3812141418457
```

```
-- 3. List all tracks along with their views and likes where official_video = TRUE.
SELECT Track,
       SUM(Views) AS total_views,
       SUM(Likes) AS total_likes
FROM spotify
WHERE official_video = 1
GROUP BY Track
```

```
-- For Top 5
SELECT TOP 5
       Track,
       SUM(Views) AS total_views,
       SUM(Likes) AS total_likes
FROM spotify
WHERE official_video = 1
GROUP BY Track
ORDER BY total_views DESC
```

```
-- Answer: 13,537 song tracks with official_video
```

```
-- 4. For each album, calculate the total views of all associated tracks.
SELECT Album, Track, SUM(Views) AS total_views
FROM spotify
GROUP BY Album, Track
Order by total_views desc
```

```
-- Answer: 18,668 albums with all associated tracks and total views
```



3

Statistical & Ranking Analytics



```
-- 1.Find the top 3 most-viewed tracks for each artist using window functions.
WITH TrackViews AS (
  SELECT
    Artist,
    Track,
    SUM(Views) AS TotalViews
  FROM spotify
  GROUP BY
    Artist, Track
),
RankedTracks AS (
  SELECT
    Artist,
    Track,
    TotalViews,
    ROW_NUMBER() OVER (
      PARTITION BY Artist
      ORDER BY TotalViews DESC
    ) AS SongRank    -- this is generated by SQL, not from your table
  FROM TrackViews
)
SELECT
  Artist,
  Track,
  TotalViews,
  SongRank
FROM RankedTracks
WHERE SongRank <= 3
ORDER BY Artist, SongRank

-- Answer: 6,214 total songs as top 3 songs for each artist
```

```
--2. Write a query to find tracks where the liveness score is above the average.
Select * from spotify
Select AVG(Liveness) from spotify  -- Average Liveness = 0.193653277666467

SELECT
  Track,
  Artist,
  Liveness
FROM spotify
WHERE Liveness > (
  SELECT AVG(Liveness)
  FROM spotify
)

-- Answer: 6,364 songs with above average liveness = 0.19
```

```
--4. Find tracks where the energy-to-liveness ratio is greater than 1.2.
SELECT
  Track,
  Artist,
  Energy,
  Liveness,
  EnergyLiveness
FROM spotify
WHERE EnergyLiveness > 1.2
ORDER BY Artist;

-- Answer: 18,799 songs with Energy-to-Liveness ratio > 1.2.
```

```
--3. Use a WITH clause to calculate the difference between the highest and lowest energy values for tracks in each album.
WITH cte AS (
  SELECT
    album,
    MAX(energy) AS highest_energy,
    MIN(energy) AS lowest_energy
  FROM spotify
  GROUP BY album
)
SELECT
  album,
  highest_energy - lowest_energy AS energy_diff
FROM cte
ORDER BY energy_diff DESC

-- Answer: 11,799 songs with energy difference.
```

```
--5. Calculate the cumulative sum of likes for tracks ordered by the number of views, using window functions.
SELECT
  Track,
  Artist,
  Views,
  Likes,
  SUM(Likes) OVER (
    PARTITION BY Artist
    ORDER BY Views
  ) AS CumulativeLikes
FROM spotify
ORDER BY Artist, Views;

-- Answer: 20,594 songs
```

--6. Query Optimization & Analysis

```
-- Query optimization = making your SQL queries run faster, use fewer resources, and scale better, without changing the results.
-- Why we optimize SQL queries
-- 1.Reduce execution time (faster results)
-- 2.Reduce CPU/memory/disk usage
-- 3.Improve scalability for large datasets
-- 4.Avoid blocking other queries in production
-- 5.Make queries more readable & maintainable

-- Sample Query:
```





Business Insights & Learnings



- Tracks with higher energy and danceability showed stronger audience engagement and streaming performance.
- Official videos significantly increased views, likes, and overall audience interaction across platforms.
- Singles consistently outperformed other album types, indicating stronger listener attention and market reach.
- A small number of top-performing tracks contributed heavily to overall artist visibility and platform engagement.
- Cross-platform analysis revealed that listener behavior differs between Spotify and YouTube consumption patterns.



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